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Paleomagnetic results from rhyolites (Early Cretaceous?) and andesite dikes at two localities in the Ocana area, northern Santander Massif, Colombia

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ABSTRACT

The mean remanent magnetization vector of six samples from andesite dikes at one locality after 15 mT ODF is D = 309, I = +77, k = 13, α_{95} = 20. Five rhyolite samples from each of two sites at a second locality yield: 50 mT ODF, D = 194, I = +30, k = 166, α_{95} = 6; 30 mT ODF D = 158, I = +46, k = 101, α_{95} = 8. These rocks are probably of Mesozoic age and the magnetization appears primary; but the vectors are unlike those for the stable South American craton at any time during the Phanerozoic. These data suggest that complex tectonic rotations have occurred in at least this part of the Santander Massif.

INTRODUCTION

In order to test the feasibility of paleomagnetic studies on rocks of the Santander Massif, suites of samples were collected at three localities in the vicinity of Ocana, Norte de Santander, in the Cordillera Oriental of Colombia. Sample localities are shown in Figure 1 (collections by Shagam, Vargas, and Rodriguez).

Locality I, at water's edge in a narrow gorge of the R. Algodonal, is a fresh outcrop of granite with a pronounced spaced (~0.5 cm) tectonic foliation. The six samples came from three (at least) thin (15–25 cm thick) closely spaced, andesitic dikes that clearly transect the foliation of the granites. Relationships between these dikes are obscured by alluvium.

Locality II consists of two sites alongside the Ocana-Abrego road (Fig. 1). The first (a), a small culvert, exposed fresh, tough, dark gray rhyolite porphyry with K-feldspar, plagioclase, and quartz phenocrysts. Samples from the second site (b) were collected from a long, high roadcut in rhyolite porphyry near the crest of a hill about 350 m southeast of the culvert. The dark fresh rhyolite produces an off-white kaolinitic regolith in contrast to the yellows of weathered granite in the area. The intimate mixture of those colors in the field underscores the problem of mapping igneous rock facies in detail, and to date most local workers group granite and volcanics in a single granite-rhyolite complex.

A few kilometres north of locality II, granite shows a highly convolute contact with upper Paleozoic (Carboniferous?) carbonates suggestive of intrusive relationships, although there is no overt sign of contact metamorphism. Localities I and II (Fig. 1)

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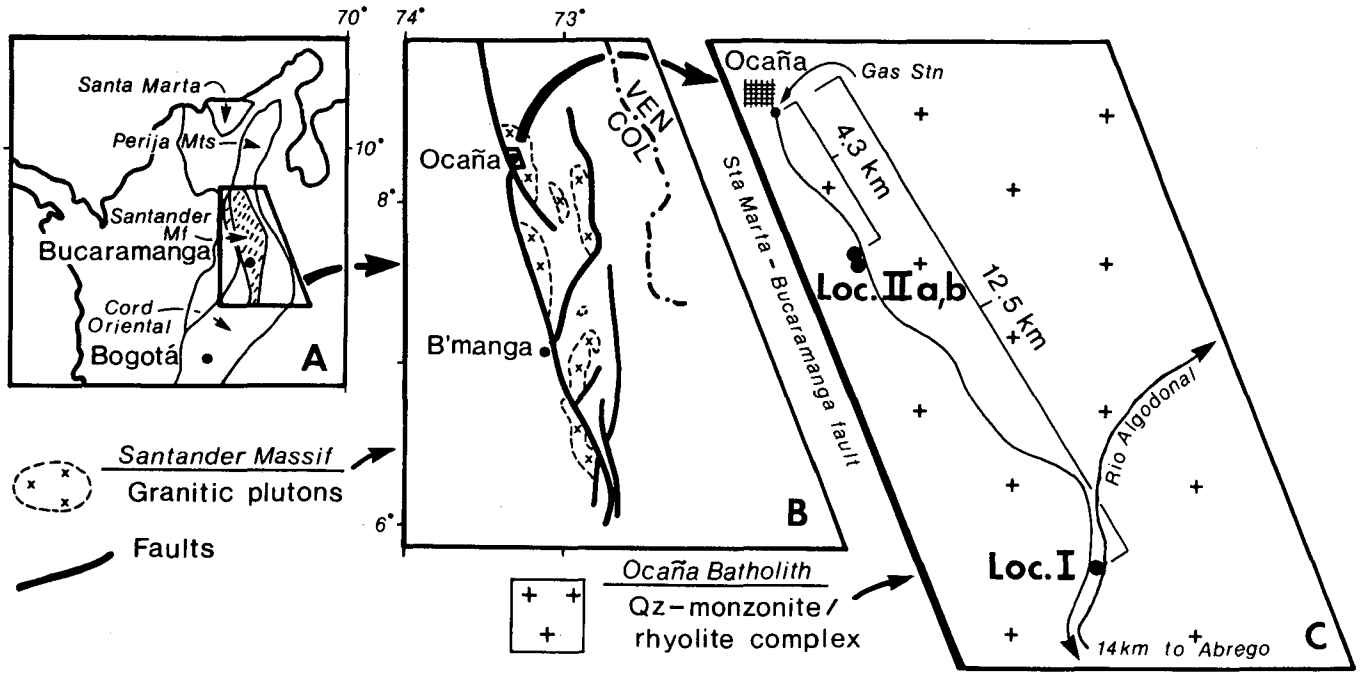


Figure 1. Location map. A = regional setting; B = Santander Massif to show distribution of granitic plutons; C = rough sketch map of part of Ocana batholith to show localities sampled for paleomagnetic study.

fall in a granite-rhyolite complex (the Ocana batholith) assigned an Early Jurassic age on the geologic map of Colombia (Arango and others, 1976). This probably derives from field observations and isotopic age determinations of Goldsmith and others (1971). They did not date the granite but inferred that it was probably of Early Jurassic age on the basis of its petrographic similarity to three other batholiths to the south which gave near-identical K-Ar (biotite) ages in the range of 192 to 196 ± 7 m.y. They did measure a K-Ar (sanidine) age of 127 ± 3 m.y. on the rhyolite near Ocana. Their location map suggests that the sample locality must have been very close to (if not the same as) our locality II roadcut. Noting that dacite porphyry dikes, ~50 to 100 km farther south, cut an Early Cretaceous formation, Goldsmith and others (1971) conceded the possibility that the rhyolite is of Early Cretaceous age. Hence, it may represent a separate phase of igneous activity distinctly younger than the granites.

The northern Santander Massif passes northeastward through a dogleg into the Perija Mountains where there is evidence for Middle Jurassic granites and a wide range of volcanic rocks, including rhyolite; these range in age from the Late Jurassic possibly into earliest Cretaceous time (see data compiled by Maze, 1984). From the foregoing, one can only conclude that the rhyolites of locality II have an age in the range of Permian to earliest Cretaceous. The pronounced tectonic foliation in the granite at locality I is puzzling; it cannot be ascribed to Tertiary shearing on the major Santa Marta–Bucaramanga fault to the west inasmuch as the andesitic dikes clearly postdate it (no Tertiary igneous activity is known on the massif). Although the regional map suggests that locality I pertains to the same granite-rhyolite complex as locality II, it may constitute an early (Permian?) phase of a lengthy igneous event. The age of the dikes relative to the rhyolites of locality II is not known.

DISCUSSION OF PALEOMAGNETIC DATA

Oriented samples were cored and their paleomagnetism measured in the Rock Magnetism Laboratory, Princeton University. The results obtained are shown in Table 1.

As is indicated by Zijderveld plots (Fig. 2a) and by the similarity between the NRM directions of individual samples both before and after demagnetization (as well as in the site means), the remanence of the andesite dikes, although weak, is approximately single component up to 20 mT, after which it

TABLE 1. MAGNETIC DATA (IN SITU) FOR SAMPLES FROM SANTANDER MASSIF

Locality/ site	Rock type	n	D_{NRM}	I_{NRM}	k	α_{95}	ODF_{mT}	D	I	k	α_{95}
I	andesitic dikes	6	342	74	10	23	15	309	77	13	20
II a	rhyolite porphyry	5	188	35	141	7	50	194	30	166	6
II b	rhyolite porphyry	5	157	49	81	9	30 50	158 154	46 54	101 26	8 15
(mean of IIa & b)		(2)	175	43	18	62	(30)	176	40	17	66

n = number of samples; D = declination, I = inclination, κ = Fisher's kappa, α_{95} = cone of 95% confidence; ODF = optimal demagnetizing field in millitesla (mT)

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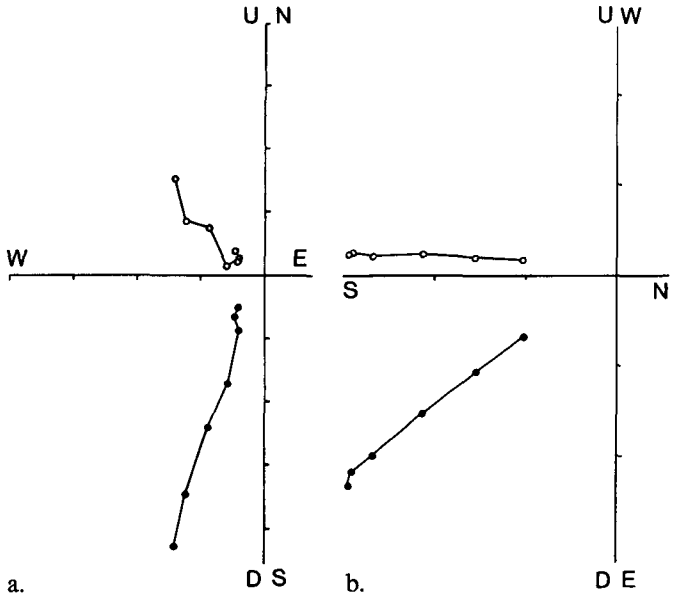


Figure 2. Representative Zijderveld plots illustrating A. F. demagnetization behavior in eight steps, N - 50 mT: a, andesite dike sample 19-2-1, units 1×10^{-3} A/m; b, rhyolite sample 20-3-1, units 1×10^{-2} A/m.

begins to scatter. These dikes are so thin that it is unlikely that secular variation is averaged out, and the scatter between samples from different dikes is attributed to this cause.

After elimination of a small soft component, the NRM of the rhyolites (sites IIa and IIb) is stronger, consistent and extremely stable to 50 mT (see Fig. 2b). The vectors for these two sites, though broadly similar, are yet significantly different; their mean is also given in Table 1.

These magnetizations are stable but clearly very different from the magnetization of the present earth's field; hence, there is no reason to suspect that the magnetizations are other than primary. The NRM directions, however, are quite unlike those for the stable South American craton at any time in the Phanerozoic. Clearly, radical tectonic rotations have occurred, but the sense and magnitude are completely unknown. In this regard it is pertinent to recall the results reported by Scott (1978) on the Payande red beds, reputedly of Triassic age. Taken in conjunction with the data from Jurassic and Cretaceous rocks farther north, as shown in Figure 3, it is clear that the aggregation of northwestern South America was a complex process (Feininger, 1983).

The data presented here demonstrate that rock types suitable for paleomagnetic study are present in the Santander Massif; a thorough study of this tectonic block, together with isotopic age dating, may prove extremely rewarding.

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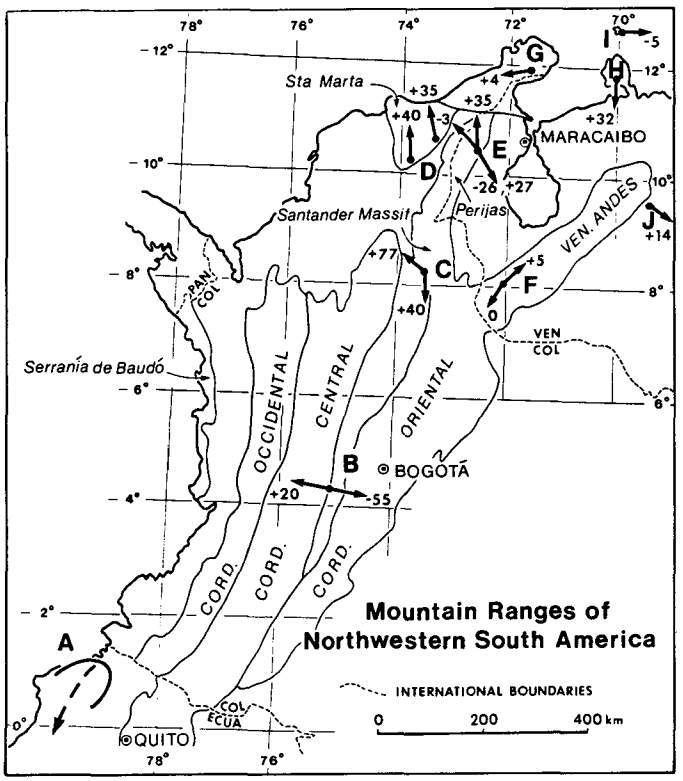


Figure 3. Northwestern South America with some geologic and paleomagnetic evidence for allochthony and major tectonic rotations. A, northern limit of allochthonous terranes in Ecuador that extend south (broken arrow) to northern Peru (Feininger, 1983). Paleomagnetic evidence of rotations and/or translations are displayed by arrows representing various generalized Mesozoic-age vectors at localities: B, Payande red beds, Scott (1978); C, this study; D, MacDonald and Opdyke (1984); E, Maze and Hargraves (1984); F, La Quinta red beds, Hargraves and Shagam (1969); G, La Teta lavas, MacDonald and Opdyke (1972); H, Santa Ana Complex, Skerlec and Hargraves (1980); I, Aruba batholith, Stearns and others (1982); J, Cerro Pelon complex, Skerlec and Hargraves (1980).

REFERENCES CITED

Arango C., J. L., Kassem B., T., and Duque-Caro, H., compilers, 1976, Mapa geologico de Colombia: Colombia Instituto Nacional de Investigaciones Geologico-Mineras, Ministerio de Minas y Energia, Bogota, scale 1:1,500,000.

Feininger, T., 1983, Allochthonous terranes in the Andes of Ecuador and northwestern Peru [abs.]: Geological Association of Canada, Abstract Volume 8, p. A22.

Goldsmith, R., Marvin, R. F., and Mehnert, H. H., 1971, Radiometric ages in the Santander Massif, Eastern cordillera, Colombia Andes: U.S. Geological Survey Professional Paper 750-D, p. 44–49.

Hargraves, R. B., and Shagam, R., 1969, Paleomagnetic study of La Quinta Formation, Venezuela; American Association of Petroleum Geologists, Bulletin, v. 53, p. 537–552.

MacDonald, W. D., and Opdyke, N. D., 1972, Tectonic rotations suggested by paleomagnetic results from Northern Colombia, South America; Journal of Geophysical Research, v. 77, p. 5720–5730.

MacDonald, W. D., and Opdyke, N. D., 1984, Preliminary paleomagnetic results from Jurassic rocks of the Santa Marta massif, Colombia, *in* Bonini, W. E.,

Downloaded from memoirs.gsapubs.org on January 18, 2014

and others, eds., The Caribbean–South American plate boundary and regional tectonics: Geological Society of America Memoir 162 (this volume).

Maze, W. B., 1984, Jurassic La Quinta Formation in the Sierra de Perija, northwestern Venezuela: geology and tectonic environment of red beds and volcanics, *in* Bonini, W. E., and others, eds., The Caribbean–South American plate boundary and regional tectonics: Geological Society of America Memoir 162 (this volume).

Maze, W. B., and Hargraves, R. B., 1984, Paleomagnetic results from the Jurassic La Quinta Formation in the Perija Range, Venezuela, and their tectonic significance, *in* Bonini, W. E., and others, ed., The Caribbean–South American plate boundary and regional tectonics: Geological Society of America Memoir 162 (this volume).

Scott, G. R., 1978, Translation of accretionary slivers: Triassic results from the Central Cordillera of Colombia [abs.]: EOS (American Geophysical Union Transactions), v. 59, p. 1058–1059.

Skerlec, G. M., and Hargraves, R. B., 1980, Tectonic significance of paleomagnetic data from northern Venezuela; Journal of Geophysical Research, v. 85, p. 5303–5315.

Stearns, C., Mauk, F. J., and Van der Voo, R., 1982, Late Cretaceous–Early Tertiary paleomagnetism of Aruba and Bonaire (Netherlands Leeward Antilles); Journal of Geophysical Research, v. 87, p. 1127–1141.

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